Supplementary Material: Retrieval-Assisted Scalar Grounding for Natural-Language Optimization: A Diagnostic Study

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This supplement preserves secondary evidence, population boundaries and source provenance. It accompanies the main article and includes the archived 217-relationship Gemini–DeepSeek–Cohere audit, with explicit recovery and denominator accounting. No new model predictions or experiments were produced for this revision.

S1 System Context and Secondary Protocol Tables

Table S1 Task-level comparison landscape for recent natural-language optimization modeling systems. This table is deliberately *not* a leaderboard: systems differ in output type, supervision, and evaluation protocol, so accuracy numbers are omitted. Abbreviations: Ret. = retrieval/memory; Sol. = solver feedback during generation or training; FT = fine-tuning or reinforcement-learning training; OR = operations research; CoT = chain of thought; Comp. = directly comparable to the InstantiationReady / Exact20 (on hits) protocol of this paper.

Method	Year	Task	Output	Ret.	Sol.	FT	Comp.
OptiMUS	2024	End-to-end (MI)LP	Model + code	no	yes	no	no
OptLLM	2024	NL→model/code	Model + code	no	yes	no*	no
Chain-of-Experts	2024	Multi-agent OR	Model + code	no	yes	no	no
LLMOPT	2025	Fine-tuned modeling	Model + code	no	yes	yes	no
OPT2CODE	2025	Retrieval + codegen	Solver code	yes	yes	no	no
ORLM	2025	Training + feedback	Model + code	no	yes	yes	no
OptMATH	2025	Data synthesis	Model + code	no	yes	yes	no
SIRL	2025	Solver-informed RL	Model + code	no	yes	yes	no
OR-R1	2026	Test-time RL	Model + code	no	yes	yes	no
DeepOR	2026	Foundation OR	Formulation	—	yes	yes	no
OptimAI	2025	Multi-agent NL–OR	Model + code	no	yes	no	no
AlphaOPT	2025	Experience library	Model + code	yes	yes	no‡	no
ORThought	2026	Logistics CoT	Model + code	no	yes	no	no
Ours	2026	Fixed-catalog inst.	Schema + scalars	yes	no	no	yes

*OptLLM supports both prompting and fine-tuned variants. ‡Library learning without weight updates. "—" denotes information not established from a verified public artifact at the time of writing. Direct comparability (Comp.) is "yes" only for systems that share our fixed-catalog scalar-instantiation task and InstantiationReady / Exact20 protocol; end-to-end solver accuracy is a different metric.

Table S2 Retrieval methods used for schema identification.

Method	Representation	Role in pipeline
BM25	Probabilistic lexical weighting	Sparse lexical baseline
TF–IDF	Weighted bag-of-words cosine similarity	Primary lexical baseline
LSA	Latent projection of TF–IDF features	Lower-dimensional lexical-semantic baseline
BGE-M3	Pretrained dense text embeddings	Modern semantic retrieval baseline

Table S3 Summary of the experimental blocks used in the paper.

Block	Primary purpose	Main outputs
Main benchmark	Retrieval and scalar grounding on NLP4LP	Schema R@1, Coverage, TypeMatch, Exact20 (on hits), InstantiationReady
Robustness analyses	Sensitivity to degraded input and lexical overlap	Orig / noisy / short retrieval comparisons, overlap-based stress tests
Numeric-extraction ablation	Isolate effect of corrected ratio-word extraction	TypeMatch and InstantiationReady changes (baseline vs. ratio-aware extraction)
Retrospective paired analysis	Quantify observed paired differences	Group-bootstrap intervals, exact swap tests and Holm adjustment
Proxy checks (60 inst.)	Coverage/type and rendered-template checks	Historical proxy and template-completion rates
Extended proxy checks (269 inst.)	Same lightweight checks on a code-available subset	Schema hit, proxy and template-completion rates
Restricted solver-backed execution (20 inst.)	Reported reference-program execution via SciPy HiGHS shim	Executable, solver-success, feasible, and objective-produced rates

S2 Secondary Retrieval and Restricted Execution Diagnostics

Table S4 Historical sanitized-retrieval aggregates (*Schema R@1*; 331 queries per cell). The 331-document diagnostic index uses 256-dimensional LSA and differs from the 335-document index in Table 1. Per-condition predictions are unavailable; agreement with the retained aggregate is not prediction-level verification.

Sanitization	TF–IDF	BM25	LSA
None (baseline)	0.9063	0.8852	0.8459
Numbers removed	0.9063	0.8973	0.8459
Stopwords removed	0.9124	0.9063	0.9184
Numbers & stopwords removed	0.9124	0.9063	0.9184

Table S5 Historically reported 60-case proxy checks using optimization-role relation repair; per-instance outputs are unavailable for reconstruction. All rates divide by 60. Proxy and template-completion predicates are defined in the text; neither checks numerical correctness.

Baseline	Schema Hit	Coverage/type proxy	Template complete
TF–IDF	0.9333	0.7500	0.7500
BM25	0.9000	0.7333	0.7333
Oracle	1.0000	0.7667	0.7833

Table S6 Historically reported gold-assisted execution on 20 instances (SciPy HiGHS MILP shim); per-instance outputs and the original manifest are unavailable for reconstruction. Every rate divides by 20.

Baseline	Executable	Solver Success	Feasible	Objective Produced
TF–IDF	0.95	0.80	0.80	0.80
Oracle	0.95	0.75	0.75	0.75

S3 Cached Grounding, References and Reproduction

S3.1 Full Factorial and Search-Retention Evidence

Section 5.5 distinguishes the historical 40-versus-7 all-key result from the controlled 49-query study. The latter's 277 scalar slots and reference-derived template inventory are fixed within each decoder comparison. The known excluded query shares a delexicalized text group with earlier training/development data; grouping uses transitive five-gram Jaccard similarity at least 0.80. Exclusion addresses that known crossing, not prior benchmark exposure or unrecognized semantic duplicates.

The design crosses two candidate constructors, three scorers, two decoders and two mention-reuse policies (24 cells). Decoder pairs share the exact same edge-score matrix, canonical lexical slot order, and zero-score null option. Scorers are an explicit ordinal encoding of the original type preference, the frozen contextual scorer, and a subtraction of only its percent/currency marker bonuses. The ordinal control introduces a documented cardinalization and does not reproduce every historical assignment. One-to-one assignment is a diagnostic restriction rather than a physical law. With unrestricted reuse and additive scores, independent per-slot maximization is already globally optimal; every corresponding local/global output agrees. Table S7 shows the one-to-one controls; all 24 cells, candidate identities, matrices and per-query predictions are retained in the reproducibility artifacts.

With contextual scores, exact assignment increases exact numerical slot counts by 30/277 (typed pool) and 33/277 (rich pool). Existing paired query-group bootstrap intervals ($B = 2000$, seed 42) are respectively $[3.14, 17.95]$ and $[4.07, 18.84]$ percentage points. These are marginal, exploratory intervals. The full-tolerance count remains

Table S7 One-to-one factorial controls on the same 49 queries and 277 gold scalar slots. Entries show greedy → exact maximum-sum assignment. E/T denote numerical exact/tolerance-correct counts. Full instances also require compatible types and every gold scalar key. These controls are distinct from the historical 331-query configurations.

Pool	Scorer	E slots	T slots	E full	T full
Typed	Ordinal	71 → 71	87 → 86	3 → 3	3 → 3
Typed	Context	106 → 136	117 → 145	3 → 4	5 → 5
Typed	No marker bonus	97 → 137	111 → 146	3 → 4	5 → 5
Rich	Ordinal	67 → 67	81 → 81	3 → 3	3 → 3
Rich	Context	100 → 133	111 → 142	3 → 4	5 → 5
Rich	No marker bonus	97 → 135	111 → 144	3 → 4	5 → 5

5/49 for both decoders, with one gained and one lost query; the paired interval is $[-6.12, 6.12]$ percentage points. A separate retrospective exact groupwise method-swap analysis covers every one-factor contrast and four numerical endpoints, using a single Holm family of 240 tests [2]. Neither contextual-score decoder contrast survives that familywise adjustment (adjusted $p = 1$). Some contextual-versus-ordinal slot comparisons survive; no whole-instance endpoint does. These results support descriptive component differences, not a confirmed whole-instance improvement or attribution of all 33 historical additional successes to search. The recorded sub-millisecond cell timings include metric and instrumentation overhead and are not clean algorithm-speed comparisons.

Candidate availability, pruning and objective quality. The earlier 50-query joint-grounding pilot selects 3 tolerance-correct instances with either local or joint scoring. In the saved common pools, 32/50 permit a criterion-correct complete mapping, but only 8/50 retain one in the top ten. Reconciliation with identical saved beam outputs separates 14 cases losing correct prefixes, 6 generating then pruning a complete correct mapping, 4 losing it at the final top-ten cut, 5 retaining it below the selected mapping, and 3 selecting it. The other 18 cases lack a complete compatible mapping in the pool or inventory. Width and ranking limitations must therefore be distinguished from candidate absence. Optimality claims about beam methods also depend on their search restriction [4]; that neural-generation result is methodological context, not a guarantee for this grounding algorithm.

An exact MILP diagnostic holds the cached local/pairwise objective and domains fixed. Among the 32 pool-compatible cases, 23 have a strictly higher-scoring incorrect optimum than their best criterion-correct mapping; 6 have tied correct and incorrect optima, and 3 select a correct optimum. The correctness-restricted solves are oracle diagnostics, never inference procedures. Exact mention-based top-ten enumeration retains a correct mapping in 9/50 cases versus 8/50 for the beam, with tied-optimum membership qualifications. More search cannot repair the 23 strict objective failures. Ten of the 50 exact top-ten lists contain duplicate numerical/type mappings realized by different mention indices. A separate 49-query follow-up ranks without evaluation

values, conditional on the same supplied inventory. It reuses 40 already-distinct lists and performs 23 additional solves for the other nine. Every retained top-ten list is certified, but criterion-correct list retention remains 9/49, with zero gained or lost queries; selected correctness remains 3/49. Thus reclaiming this alias budget supplies no observed correctness gain and no justification for full-dataset expansion. Mention equality, numerical equality and semantic identity are different concepts. No gold value is used to rank inference candidates.

Detailed source/reference conventions, the versioned correction overlay and the historical tolerance limitation are given in Appendix S3.2. No reference correction is used during inference, and no new hard unit or mathematical relation was validated by these studies.

S3.2 Reference Conventions and Correction-Only Reanalysis

All 12 previously flagged percentage slots were inspected against the original query, parameter descriptions, schema equations and reference-code annotations. A separate versioned overlay, `numerical_reference_v2`, contains ten fractional-reference corrections (Table S8). The literal percentage and a direct share-coefficient equation establish each conversion: a share p in $X \geq p(X + Y)$ or its rearrangement $(1 - p)X \geq pY$ must use the fraction, not percentage points. Upper bounds reverse the inequality. These corrections do not alter original reference files, any historical measurements, or inference predictions. They establish a source/schema-consistent convention, not dataset-maintainer approval or correctness of every other constraint in an instance.

Table S8 Complete fractional-reference change inventory. IDs abbreviate `nlp4lp_test_ID`; all values are scalar. Every entry has full source text, a literal span, equation/code witnesses, hashes and a confirmed evidence classification in the overlay.

ID	Parameter	Original	Fractional reference
48	MinLycheePercentage	40	0.40
78	MinChickenCurryPercentage	25	0.25
102	MinPercentageAdultVaccines	70	0.70
157	MaxRickshawPercentage	40	0.40
170	MinBoatTripPercentage	50	0.50
174	MaxPercentageLabradors	60	0.60
177	MaxCompanyCarRidePercentage	60	0.60
180	MinCarTripPercentage	30	0.30
187	MaxVanPercentage	30	0.30
215	MaxTrolleyTransportPercentage	40	0.40

The two remaining flags are the two snack-mix parameters in query 206. They retain 20 and 35 as percentage points: the complementary-ingredient constraint explicitly uses

$(100 - p)/100$. Converting those references alone would break that equation. Instead, a separate formulation overlay inserts the missing divisions in the other ingredient constraint:

$$(p_1/100)x_1 + (p_2/100)x_2 \leq b_{\text{cat}}, \tag{S1}$$

where $p_1 = 20$, $p_2 = 35$, x_1, x_2 are mix masses in kilograms, and $b_{\text{cat}} = 20$ kilograms is the available cat-paw ingredient. This is an algebraic/dimensional correction only; no model is executed. Thus a percent-named parameter does not imply a universal fractional reference representation. Query 102 additionally has a metadata-key alias, and query 177 a pluralized coefficient identifier in another constraint; the numerical overlay does not repair those independent schema defects.

Reanalysis is restricted to predictions actually saved. None of the ten changed-reference cases is in the 49-query primary factorial, so all of its metrics and the 240-test family are unchanged. Only development case 177 has affected factorial predictions; the existing version-1 sensitivity is reusable because its correction is unchanged. It adds no fully correct development instance. Reference-only cases without predictions are not run.

For the distinct, pre-ratio Stage-A cache, saved selections over all 331 queries and 1,847 gold scalar slots permit a separate correction-only comparison: numerical exact slots change from 348 to 356, tolerance-correct slots from 401 to 409, and full type-compatible tolerance instances remain 6/331. Missing saved selections count wrong. These are not replacements for the headline typed-greedy 7/331 or constrained 40/331 results, whose historical per-slot values are unavailable. The Stage-A counts use actual saved selections; an earlier metadata-filtered error partition counted 394 correct slots because seven otherwise correct saved selections lack slots in the later metadata-only snapshot. The inventory difference is explicitly reconciled, rather than described as an experimental change.

The historical numerical tolerance remains $|\hat{y} - y|/\max(1, |y|) \leq 0.20$: for fractions it is absolute 0.20, so prediction 0.40 passes against reference 0.25. It is not uniformly a 20% relative tolerance. The pilot's numerical exact criterion uses the same scaling with threshold 10^{-9}, not bitwise equality. Slot-level numeric correctness does not impose a type gate; full-instance correctness also requires the schema, all gold keys and compatible types. Candidate-pool coverage instead asks whether any permitted option could satisfy a criterion, without claiming that inference retained or selected it. No new tolerance is chosen and no gold value is used during selection.

S3.3 Populations, Evidence and Reproduction Boundaries

The completed archival investigation found no authenticated headline scalar-assignment export. The four existing metric CSVs byte-match their first committed versions; this establishes stored-record integrity, not independent reevaluation of the values originally selected. The inspected original runners serialized 13 metric columns and omitted scalar mappings. Their absence is not evidence of deletion, and source code alone cannot reconstruct a historical prediction.

We distinguish five evidence types. *Original aggregates/query metrics* support summary-statistic reconstruction but not recovery of scalar choices. *Preserved predictions* include top-1 schema IDs, Stage-A selections and later cached mappings.

Deterministically reconstructed statistics include residual partitions and paired tests derived from those existing records. *Historical diagnostic examples* document selected cases or separate replay configurations; they are not complete original samples. *Unmatched archival candidates* have no authenticated run identity and are excluded from evaluation. No newer outputs replace missing predictions.

Table S9 Evaluation populations and preserved evidence. These populations and configuration versions must not be interchanged.

Evidence	Unit and denominator	Available observations
Retrieval	331 queries per variant	Top-1 schema IDs
Main scalar diagnostics	331 queries per arm	Query metrics, not slots
Conditional Exact20	291/291/303/320 queries	Per-query fractions
Pre-ratio Stage A	1,705 rows / 311 queries	Mentions and selections
Joint pilot	50 queries; 280 gold scalars	300 method/query traces
Primary factorial	49 queries; 277 gold scalars	Choices and matrices
Initial factorial collection	64 cases; mixed populations	64 checkpoints
Distinct-class enumeration	49 primary queries	49 saved checkpoints
Subset proxy/solver checks	60/269/20 filtered queries	Reports/aggregates
External generation	Same 18 source problems	Artifacts/execution logs
External extraction	1,000 instances	Extracted sets and counts
Relationship QC	217 items / 16 problems	651 final objects and recovery lineage

The Stage-A cache covers 311 queries with 1,705 slot rows; its separate full-frame sensitivity uses 331 queries and 1,847 gold scalars, counting missing selections as missing. The initial factorial collection contains 49 primary, 12 development, one overlap-reconciliation and two reference-only cases. All 1,176 primary method/query choices were reevaluated from saved indices/candidate values with historical type rules; no decoder or solver was rerun.

The headline per-query files are under `results/selective_grounding_rerank/`, `results/final_resubmission_method/`, `results/dense_retrieval_bge_m3/` and `results/oracle_recomputation_2026-08-15/`. Matched alternatives are under `results/stage6_matched_grounding_baselines_2026-08-27/`; joint traces are under `results/joint_grounding_2026-10-27/`. Original factorial and enumeration

checkpoints are under `results/wulver_grounding_attribution_2026-10-27/slurm_1462366/` and `results/grounding_followup_2026-10-27/slurm_1462557/`.

Saved top-1 IDs reconstruct retrieval rates; saved query metrics reconstruct the main diagnostic tables and paired statistics (10,000 group resamples, NumPy seed 42). The corrected TF–IDF variant mean is 0.8640, computed before rounding from $(301 + 299 + 258)/(3 \cdot 331)$. Main scalar files cannot support corrected-reference reevaluation. The BGE reporting generator writes literal metadata/aggregates, not a runtime receipt: its revision field and inferred device do not authenticate executed weights, environment or hash-dependent slot order.

The sanitized-retrieval and 60/269/20 complete results remain supported by historical aggregates/reports only. Five reported examples each from 60 and 269 include three shared query IDs, but do not recover either full manifest. The 20-case outcome writer itself omits full scalar maps. Imported overlap-study hit flags, metadata-order diagnostic replays and unauthenticated loose blobs cannot substitute for the missing target outputs.

External generation preserves artifacts and execution records, but incompletely retained prompts prevent authentication of every historical input. Gold objectives are scoring-only in inspected generation paths; objective agreement is not formulation equivalence. The extraction audit preserves automatic sets/counts. Its later corrected aggregate ratio is checkable, but missing corrected per-instance counts prevent independently reconstructing its historical bootstrap interval.

The M3 correction uses `tools/reaggregate_common18_m3.py` and the pure `baselines/comparison/objective_evidence.py` module. From a checkout containing the M3 correction scripts, run `python` with script argument `tools/reaggregate_common18_m3.py`, `--output` and a separate local audit directory. The command reads saved records only and never overwrites historical results. Independent verification and regression instructions accompany the local correction ledger under `docs/manuscript_m3_2026-10-28/`. This ledger identifies every case, source hash, historical/corrected disposition and precision exclusion.

The M5 local release candidate supplies a machine-readable file inventory, hashes, excluded-file reasons and offline commands in `reproducibility/m5/README.md`. Its deterministic entry point, `tools/verify_llm217_m5.py`, reconstructs judgments from frozen HTTP responses without importing historical validators or model SDKs. The 651-row judgment ledger and the 20 invalid-case IDs are derived exports; originals are preserved. Account/project identifiers are removed only from explicitly marked candidate release copies, with separate original and release hashes. No raw NLP4LP cache or OptMATH problem corpus is redistributed. The already public derived-metrics dataset keeps its existing CC BY-NC-SA 4.0 license; NL4Opt source material keeps its MIT notice. M5 is a local author-review candidate, not an uploaded release.

Direct offline checks reproduce retained metric aggregates, corrected common-18 text measurements and relationship validity/agreement. Fresh source-level evaluation requires separately obtained upstream data and appropriate dependencies; it was not run here. Historical aggregates without membership records and missing original headline scalar maps remain verifiable only to their retained evidence boundaries. No reconstructed output fills these gaps.

S3.4 Input Dependencies and Identifiable Conclusions

Table S10 Verified input boundaries. "Reference-derived" describes provenance even when a cached template is treated as a supplied catalog.

Stage	Prediction-time inputs	Reference use / interpretation
Retrieval	Query; reference-derived catalog descriptions	Partially assisted known-template identification; target ID used only for scoring, except Oracle
Main grounding	Query; selected template keys; query gold scalar eligibility	Gold-assisted assignment; gold magnitudes used only for scoring
Joint/factorial controls	Cached mentions; template names, shapes, descriptions and available relations; name-inferred types	Partially assisted; no query gold-key filter or value-guided inference ranking
Reference execution	Selected reference program; default parameters; predicted scalar replacements	Gold-assisted; structured and unfilled values retained
External generators	Problem text; fixed prompt; model; own solver feedback where configured	Inspected prompts exclude reference targets; gold objectives score outputs

For the headline scalar criterion, let F be the set passing schema match, full eligible coverage, all types and full numerical tolerance; let S and I pass Strict and ordinary coarse readiness. By their definitions, $F \subseteq S \subseteq I$: full coverage/type gates imply the 0.8 gates, and Strict adds schema match. Thus the saved ratio-aware counts satisfy $7 \leq 255 \leq 265$ on the same 331-query frame. This is a logical relation between conditional diagnostics, not a bound on autonomous or semantic accuracy. Without assumptions linking a system supplied with a reference inventory to a system that must infer that inventory, the latter's success rate is unidentified. Nor does stored-reference tolerance certify source semantics when the references themselves contain errors.

The absence of new fitting in this revision does not establish evaluation independence. The earlier learned train/development/test subsets all come from the 331-query evaluation-origin corpus; retained grouping records identify train/development and development/test crossings. The 49-case primary slice excludes the known query-291 crossing, but does not certify new template families or a fresh test set. The separate 399-problem NL4Opt quarantine is not an NLP4LP split and was not applied to these tables.

S3.5 Provisional Availability and Outstanding Releases

The public Hugging Face metrics are the historical v1.2 dataset at revision `24a047fa5d661be6cfcf75980ac8f62e6c942243`. They are not an uploaded M5 release. The public GitHub repository retains historical records, including comparison fields superseded by the documented terminal-objective correction. The Phase 2 branch publishes manuscript sources, supplementary summaries and deterministic aggregate checks; it does not publish the local M3/M5 evidence bundles or private author-review records.

Final source-level reproducibility would benefit from a separately reviewed additive release of the corrected common-18 ledger/evaluator, permitted relationship prompts and responses with recovery lineage, their independent validators, and the nonprivate D1 case/reference package. Such a release requires its own artifact and access review. It cannot recover missing headline scalar mappings, unavailable corrected extraction rows, full-precision PaMOP returns or incomplete historical environments and prompts. Upstream access conditions continue to apply; no dataset or Hugging Face record was changed for this manuscript revision.

S3.6 Historical Supervision Experiments: Evidence and Limits

Completed training, negative performance, inconclusive evidence and unavailable supervision are different outcomes. Retained training logs establish that a DistilRoBERTa mention–slot classifier was actually fitted, rather than merely implemented. Its weak labels selected the first numerically matching mention. The largest retained run (job 854626) used 230/50/50 internal train/development/test instances drawn from the existing NLP4LP evaluation corpus. The logged selection counts are 51/280 slots and 0/45 legacy exact-instance outcomes. Five nominal test instances do not enter that latter denominator. Those metrics are not the all-required-scalar criterion used in this manuscript.

A later summary's claim that this run was worse than a rule baseline on every metric conflicts with the preserved stdout. The audit also identifies an evaluator path that can instantiate an untrained neural head for a nominal rule configuration. Missing original run checkpoints, exact run source SHAs and prediction traces prevent resolving that comparison by replay. Therefore these logs demonstrate completed weak-label learning with poor recorded absolute outcomes, but not a clean causal comparison of neural learning against rules. No headline result is substituted from the conflicting summary.

P0 is a separate completed classifier-plus-assignment study. Logistic regression over 25 handcrafted features and one frozen MiniLM cosine feature was selected on development slot selection; the encoder was not fine-tuned. On its 50 oracle-schema cases, independent selection has coarse readiness 0.80, exact maximum-weight assignment 0.78, and assignment plus repair 0.78, versus a separately retained typed-greedy comparator at 0.86. These are the saved readiness outcomes, not whole-instance numerical correctness. The probability outputs have not been shown calibrated; the exact assignment has no private null option, and several components differ from the comparator. Its first-value-match labels and evaluation-origin split do not establish an independent semantic test.

Both histories used internal subsets of the 331-query corpus; disjoint example IDs alone did not establish independence. The saved text-group audit finds train/development and development/test overlap. This compromises any later use of those fits as independent evidence on the full 331 queries. It does not retroactively turn every internal observation into a nonexistent experiment. The defensible lesson is narrower: fitting an equality-derived target and optimizing its scores did not establish trustworthy occurrence-specific supervision.

MiniLM cross-encoder and multitask grounder implementations also exist, but a completed non-smoke run for the former and a completed NL4Opt auxiliary neural checkpoint for the latter were not verified. They are not counted as negative model results. Actual GAMS and targeted-synthetic auxiliary runs used only two and one effective test problems; they are too small and weakly supervised to justify general transfer claims. Those execution records remain in the audit inventory rather than a new performance table. Fine-tuned retrieval addresses a different task, and an encoder substitution alone is not a new relationship-grounding contribution.

S3.7 NL4Opt Relationship and Independence Audit

The audit reads 713 NL4Opt source problems at pinned revision `49f1e0d66b7fdcd33305a7f281c2a7c13f5620ea`. Its 3,114 positive proposals and 3,246 ambiguous links are derived occurrence–declaration candidates. They do not constitute new independent human annotations. A conservative source-template investigation propagates quarantine to 399 problems; 314 remain unflagged but uncertified. Earlier lexical-screen negatives and later structural-template concerns are compatible because they test different relations. Quarantine counts are precautionary grouping outcomes, not measured contamination or semantic-error rates.

A separate relationship-labeling recovery audit is discussed in the main paper. Its numerical provider comparison, recovery protocol and complete-case denominators appear in main Section 5.8.3 and Online Resource 1, Section S6.1. No machine consensus has been accepted as a human semantic reference, and no model is trained from that packet for this revision.

S4 LLM Formulation Provenance and Individual Outcomes

S4.1 Implementation Provenance and Corrected Predicates

All four evaluated systems attempt the same 18 source problem numbers, selected before the comparison by the fixed `comparison-nlp4lp-common-18-v1` manifest. These numbers map to zero-based NLP4LP query IDs; retained input hashes agree for every Python system. Full rendered prompt retention is incomplete. No model inference or solver execution is repeated in this correction.

The Azure responses identify `gpt-5.4-2026-03-05`. OptMATH checkpoint revision is `617fe77`; upstream code is `f15bbc4477c70db85ad148df8bcc1b780bca0f8c`. ORLM checkpoint revision is `94fdc3c5738c6536d4880dc19a78f215529181c5`; upstream code is `33bc47d0a1d1710d24ab839118bdf4cb89b9e31b`. Generation adaptations

Table S11 External-baseline provenance on the common-18 subset. The PaMOP row is a project reconstruction; adapted rows use released checkpoints with project-maintained interfaces. Ours is listed for context only; native metrics are not comparable to Table 8.

System	Provenance	Output representation	Solver environment / status	Cases
Ours (this paper)	NATIVE	Scalar slot assignments on retrieved schema	No solver call in core benchmark	18
PaMOP [6]	Project reconstruction	AMPL formulation + solver code	Local AMPL/Gurobi; 13/18 successful	18
ORLM [3]	ADAPTED-OFFICIAL	LP formulation + solver code	Solver dependency (`coptpy`) unavailable; execution blocked	18
OptMATH [5]	ADAPTED-OFFICIAL	LP formulation + solver code	Gurobi (`gurobipy`); 15/18 executed	18
Generic LLM (GPT-5.4-2026-03-05)	GENERIC-ZERO-SHOT	Gurobi solver code, zero-shot	Gurobi (`gurobipy`); 16/18 executed	18
DeepOR [7]	—	Formulation (paper-described)	No evaluable artifact located in the August 2026 audit	0
OR-R1 [1]	—	Formulation + solver code	Official code found; checkpoint unavailable in the August 2026 audit	0

identify local `69df7c030fabb3bd785684b68a844929487603b9`, except ORLM at `1831396b0b3d4428415e354b0a4e1fcbc658df26`. The later direct-Gurobi execution script is at `0197f0cf737f162bd65128c532c708177f475e33`; it is absent from the earlier generation commit. These stages have separate provenance. Successful logs name Gurobi 13.0.2; 12 PaMOP successes name that solver and one is solved by AMPL presolve. These records do not certify every environment default or hosted weight byte.

S4.2 Terminal-Objective Extraction and Precision

The original Python execution harness selected the first objective-like string, which could be a heuristic incumbent. The correction instead requires a terminal optimal solver summary and checks it against the final standalone objective printed by the generated program. The original generated code, stdout and reference objectives are preserved; no program or model is rerun. Nine stored objective fields differ from these terminal values, and seven agreement labels change. All 31 successful Python runs have a valid terminal/printed objective pair. The corrected agreement counts are 11/15 for adapted OptMATH and 12/16 for zero-shot GPT-5.4, with four disagreements in each conditional set.

Table S12 Actual historical implementation and configuration, not original-author performance reproduction.

System	Model/implementation	Settings and actual solver
PaMOP	Project-maintained reconstruction; Azure GPT-5.4	$T = 0.2$; decomposition; at most 5 correction iterations; AMPL/Gurobi
Zero-shot	Azure GPT-5.4 fixed prompt	$T = 0$; one generation; 8192-token cap; gurobipy
OptMATH	Adapted released Qwen2.5-7B checkpoint	$T = 0.8$; sampling; seed 0; 8192-token cap; gurobipy
ORLM	Adapted released LLaMA-3-8B checkpoint	Greedy; seed 0; 8192-token cap; coptpy unavailable

For continuity the reported Python proxy retains its original predicate. Let $a = \mathrm{round}_{\mathrm{even}}(\hat{z})$ and $b = \mathrm{round}_{\mathrm{even}}(z^*)$, using Python's float conversion and nearest-integer ties-to-even behavior. Agreement is

$$\mathbf{1}\left[\begin{cases} |a-b| \le 0.05, & b = 0, \\ |a-b|/|b| \le 0.05, & b \neq 0. \end{cases}\right]. \tag{S2}$$

Full available numerical tokens are retained before that explicit comparison step. A separately recorded unrounded sensitivity uses absolute 0.05 at a zero reference and relative 0.05 otherwise; it gives identical classifications on these 31 cases. This agreement does not validate rounding for other objectives or make the systems' protocols identical.

PaMOP's historical absolute-10^{-6} objective comparison has a different precision problem. Common-18 case 254 displays 16666.7 although its solver summary prints 16666.66667 against reference 16666.666666666664. Even the latter text cannot certify that tolerance. More generally, the retained displayed equality does not establish an unrounded return at 10^{-6}. We therefore withhold its aggregate objective-agreement score. Among 11 executed cases with numerical references, two discrepancies exceed a conservative last-displayed-unit uncertainty bound and nine remain precision-unresolved; two other executed cases have no reference objective. An algebraic witness for case 254 is documented separately and is not substituted as a recovered numerical solver result. No revised exact-agreement fraction is inferred.

Five PaMOP cases finish with parse failure; nine of its 13 successful cases execute without correction and four after correction. Adapted OptMATH has three unsuccessful processes, and zero-shot has two. Four of these five Python failures (cases 219 and 237 for both systems) invoke Gurobi, terminate infeasible or infeasible-or-unbounded, and then fail accessing an objective; OptMATH case 232 has a distinct nonlinear-constraint API construction error. These stages are not interchangeable semantic diagnoses. All successful runs report optimal/solved generated-model outcomes, with conditional feasibility counts 13/13, 15/15 and 16/16 for PaMOP, OptMATH and zero-shot, respectively. This does not show that the generated models express the source problems.

ORLM has code extracted on all 18 cases but no execution because `coptpy` is unavailable. Its execution and objective columns remain N/A. Table S14 gives each case's outcome, and Table S12 records configurations. Missing rendered historical prompts and incompletely authenticated environments still limit exact generation replay. Objective agreement and successful execution do not establish formulation equivalence; the core scalar-grounding method remains outside these columns.

Table S13 Noninterchangeable failure and assessment stages from archived records.

System	Stage/outcome	Count
PaMOP	Successful without / after correction	9 / 4
PaMOP	Final parse failure	5
PaMOP	Executed: robust discrepancy / precision unresolved / no reference	2 / 9 / 2
OptMATH	Unsuccessful process / objective disagreement among comparable executions	3 / 4
Zero-shot	Unsuccessful process / objective disagreement among comparable executions	2 / 4
ORLM	Execution environment unavailable	18

S4.3 Per-Case Outcomes and Precision Boundaries

P denotes final parse failure, E an unsuccessful process, N an unavailable execution environment, Y+ a classifiable objective match, Y– a classifiable discrepancy, Y? a successful run without a numerical reference, and U an unresolved absolute-precision comparison. U is neither a failed model nor a confirmed agreement. The historical five false feasibility flags on failed Python processes are excluded. Each source record and its hash is identified in the local versioned correction ledger.

For PaMOP case 254 the retained generated LP minimizes $100x + 72y$ subject to $60x + 40y \geq 10000$, $x \geq (x + y)/3$ and nonnegative variables. Since $100x + 72y \geq (5/3)(60x + 40y) \geq 50000/3$, with equality at $x = 500/3, y = 0$, the mathematical model has objective $50000/3$. This algebraic witness explains the display-rounding artifact but is not a recovered full-precision historical solver return or a complete source-semantic validation. The printed summary 16666.66667 is still insufficient at absolute 10^{-6}, so the case remains U in the machine-measurement ledger.

The archived comparison contains no evaluated common-18 generation for DeepOR or OR-R1. Their historical August 2026 availability context is retained, with no fabricated zero accuracy. Different prompts, adaptations, sampling and correction budgets, reference predicates and execution availability prevent a provider/model ranking. No scalar-only readiness score is entered as executable-formulation performance.

Table S14 Common-18 corrected dispositions from archived records; no regenerated programs or solves.

Problem ID	PaMOP	OptMATH	ORLM	Zero-shot
14	Y–	Y+	N	Y+
23	U	Y+	N	Y+
34	U	Y+	N	Y+
59	U	Y–	N	Y–
69	U	Y–	N	Y–
72	U	Y+	N	Y+
84	P	Y+	N	Y+
88	U	Y+	N	Y+
96	Y–	Y–	N	Y–
117	P	Y–	N	Y+
190	P	Y+	N	Y+
202	U	Y+	N	Y+
208	U	Y+	N	Y+
219	Y?	E	N	E
232	P	E	N	Y–
237	Y?	E	N	E
254	U	Y+	N	Y+
262	P	Y+	N	Y+

S5 Closed Slot-Participation Audit

The G1–G3 audits were bounded source/record investigations and produced no experimental predictions. G1 found missing validated autonomous slot discovery; G2 distinguished slot participation from mention-candidate filtering. For G3, let $P_{\text{scalar}}(\hat{s})$ select only metadata entries whose declared shape is empty. It compares $K_G \cap P_{\text{scalar}}$ against P_{scalar} using saved TF–IDF IDs, without running a decoder.

The original implementation intersects K_G with all metadata keys, falling back to reference parameter keys; it does not first restrict shapes. Its saved slot counts sum to 1705, whereas strict metadata scalars give 1675, differing by 30 slots in seven cases. These baseline definitions cannot be interchanged. The changed cases add 134 occurrences: 132 keys are unannotated in the query reference, and two have symbolic string values. Their correctness is unknown, not automatically false. Whole-key success cannot distinguish the two conditions in 29 changed cases because both lack required keys; the remaining case cannot certify numeric scalar success. Equal-set cases supply no intervention under identical deterministic ordering. This is structural identifiability, not a measured zero effect or evidence of equivalence.

Only query 102 is both a changed case, a schema hit and a source of shared numeric targets (three). Its metadata/reference names differ and existing percentage conventions are inconsistent. Wrong-schema shared names do not certify shared semantics. Added

Table S15 Static G3 inventory accounting on 331 saved TF–IDF records; these are set counts, not performance outcomes.

Measure	Count
Equal /different participation sets	301 /30
Additional schema-slot occurrences	134
Gold slots absent from predicted scalar metadata	172 /1847
Changed cases missing required gold keys in both conditions	29
Changed case without numeric scalar references	1
Cases where strict metadata-scalar A differs from original fallback A	7

slots could compete for mentions under no reuse, but that mechanical possibility does not supply an informative performance study here. All 49 prior primary factorial cases have equal sets. No 10–20-case preflight or new 331-case run was executed or recommended. The closed audit preserves per-case inventories, hashes and exact denominators; no algorithm or reference repair was applied.

S6 Relationship-Labeling Evidence and Recovery

S6.1 Provenance, Requests and Denominators

The packet contains 217 unique occurrence–declaration pairs in 16 source problems, all from NL4Opt train (713 source records), pinned to revision `49f1e0d66b7fdcd3 3305a7f281c2a7c13f5620ea`. The complete source-file SHA256 is recorded with the source-to-result manifest. Selection is a high-priority convenience handoff with category coverage from an exposed 100-problem review packet, not a new probability sample or 217 independently sampled observations. Relationship IDs, problem IDs, source hashes and every selected object are retained. Reused mentions or declarations remain separate relationships. There are no missing or duplicate targets in the original 217-target frame for any model; historical replacements are additional attempts at a subset, not additional independent observations.

Each service originally received 33 batches of up to eight targets, with complete source text and supplied variables, objective and constraints; peer judgments were withheld. After offline repair, 92 targets remained invalid and were historically reissued in smaller batches with explicit identifier and quotation instructions. Cohere additionally received a native JSON Schema with identifier enums. These changes affect treatment comparability.

All retained study calls are dated October 27, 2026 (UTC). Original raw-response timestamps span 20:58:09–21:12:23; targeted-recovery responses span 21:45:10–21:52:12. Synthetic access probes and model-list lookups in preflight are not study judgments. Vertex uses the v1 `generateContent` route (project ID removed from release copies), io.net uses https://api.intelligence.io.solutions/api/v1/chat/completions, and Cohere uses https://api.cohere.com/v2/chat. The requested models are `gemini-3.1-pro-pre view` through Google Vertex AI, `deepseek-ai/DeepSeek-V4-Pro-0813` through io.net

io.intelligence (recorded FP8), and `command-a-reasoning-08-2025` through Cohere's direct v2 API. DeepSeek here names the served model, not a direct DeepSeek API. All use temperature 0 and a 12,000-output-token cap; Gemini uses LOW thinking, Cohere a 256-token thinking budget, and io.net its default reasoning behavior. Sampling/analysis seed 20261027 is not an inference seed. The identifiers authenticate the requested or echoed service names, not underlying weight checksums or every server default. Gemini and io.net responses echo their requested model strings; Cohere does not supply a separately authenticated version in the retained response. No serving weight checksum is available. Unrecorded top-p, inference seeds and other defaults are unknown.

Table S16 Retained HTTP attempts, distinct from model-item judgments. A response with HTTP 200 may still contain invalid judgment objects. The initial Cohere recovery transport was rejected before the amended native-schema transport.

Service	Original attempts	Original HTTP 200	Recovery attempts	Recovery HTTP 200
Gemini	33	33	0	0
io.net/DeepSeek	33	33	3	3
Cohere	33	32	58	27

The remaining original Cohere attempt returns HTTP 422. Recovery retains 27 Cohere HTTP 400 responses from the initial native-schema format, four HTTP 422 responses, and 27 HTTP 200 responses after amendment. These 58 attempts must not be described as 27 total requests. The completed replacement batches target four io.net items and 88 Cohere items. Neither HTTP retries nor transport amendments constitute new target observations. The amended schema removes unsupported native length constraints and supplies explicit enum types; local validation is unchanged. This is a historical protocol deviation, fully retained, not an M5 rerun.

The original response contract requires exactly one object per requested ID and ten named fields: item ID, decision, explanation, evidence, variable IDs, constraint IDs, missing relations, missing implicit objective terms, shared-mention ambiguity and uncertainty. Evidence contains literal contiguous source quotes and supporting/contradicting/context tags. A unique complete envelope may be extracted from prose, an alias may be mapped only via the source's explicit alias dictionary, and structured omission-note objects may be serialized with an exact JSON round trip. No decision or quotation is rewritten. These deterministic rules repair 59 originally invalid objects; 500 originally valid objects remain unchanged. The final set adds 72 valid historical replacements and retains 20 invalid ones. Structurally invalid labels are not counted as a fifth nominal category or scored as semantic errors.

Table S17 Valid-label distributions after Recovery v2.1. C = CORRECT, I = INCORRECT, A = AMBIGUOUS, CD = CANNOT_DETERMINE; these are model assertions, not human reference labels. Invalid judgments are excluded from these label totals.

Service	C	I	A	CD	Valid total
Gemini	134	82	0	1	217
io.net/DeepSeek	130	84	2	1	217
Cohere	177	20	0	0	197

For $N = 197$ complete items and $n = 3$ raters, let n_{ic} count judgments in nominal category c. We recompute

$$\bar{P} = \frac{1}{N}\sum_i \frac{\sum_c n_{ic}(n_{ic}-1)}{n(n-1)}, \quad P_e = \sum_c \left(\frac{\sum_i n_{ic}}{Nn}\right)^2, \quad \kappa = \frac{\bar{P}-P_e}{1-P_e}. \tag{S3}$$

Using all four categories in Eq. (S3) gives $\bar{P} = 0.7394247039$, $P_e = 0.5858578050$ and $\kappa = 0.3708071786$. Complete-case selection is conditional on output validity and may be informative. Offline-only repair yields 128 complete items and $\kappa = 0.4293652076$; the original 89 complete items give 0.5016425265, unchanged when those exact 89 items are assessed in the final records. No interval or inferential model treats the 217 relationships as independent. The residual 20 items are all final Cohere nonliteral-evidence failures, individually listed in the machine-readable `llm217_missing20.csv`. The complete-case partition $119 + 77 + 1 = 197$, together with these 20, exhausts the frame of 217.

Among jointly valid pairs, nominal agreement is Gemini–DeepSeek 213/217 (98.16%), Gemini–Cohere 123/197 (62.44%), and DeepSeek–Cohere 120/197 (60.91%). The complete three-model denominator is 197, comprising 119 determinate unanimous cases, 77 disagreements and one unresolved case. The remaining 20 of 217 are invalid/incomplete. The unresolved case has three `CORRECT` labels but no verified lexical endpoint, so raw label unanimity is 120/197, while determinate unanimity is 119/197. Neither is semantic accuracy.

The complete-case category marginals are 420 CORRECT, 168 INCORRECT, two CANNOT_DETERMINE and one AMBIGUOUS, totaling 591 judgments. No population confidence interval, significance claim or model ranking is inferred for these selected, clustered items.

S6.2 Identified Protocol and Disagreement Witnesses

These examples are traceable observations, not human reference adjudications. Item `00330c75d90c5e1b24b115da`, problem −790646361, originally uses Cohere variable ID `spoon` where the supplied variable is `spoons`. The explicit source alias permits deterministic repair without changing the decision. Item `32f85abf70d7e3065d1ef01f`, problem −2101760318, retains a Cohere quote “You must consume at least 50 units of

sugar" that is not a literal substring of that problem's source. It remains invalid; the numerical decision is not adjudicated by this string check.

For item `a8ccbba13aa191a89f74a58b`, problem −790646361, the three valid labels are CORRECT/INCORRECT/CORRECT (Gemini/io.net/Cohere). The explanations conflict over whether the targeted repeated 1.5 occurrence is a steel or rubber quantity. Item `62b5c72b04145f4b57de8ef9`, problem 1470713712, has the same label pattern, with competing explanations about the 40-pound resource bound and the supplied coefficients. These are examples of equal-value/resource and declaration-scope disagreement, not evidence that one service is generally right. Item `9daac1d92d950356baeeb7ed`, problem −1202959445, has three CORRECT labels but no lexical occurrence endpoint. The protocol classifies it UNRESOLVED, explaining why raw label unanimity (120) exceeds determinate unanimity (119).

Table S18 Per-problem relationship counts, complete three-model cases and final dispositions. These 16 clusters, rather than 217 independent problems, define the source grouping. U denotes determinate unanimity, D disagreement, R unresolved and I invalid/incomplete.

Problem ID	Items	Complete	U	D	R	I
-1045601007	13	12	10	2	0	1
-1119351614	13	13	10	3	0	0
-1202959445	6	6	5	0	1	0
-1734964325	1	1	0	1	0	0
-1761770632	14	12	7	5	0	2
-2101760318	16	15	8	7	0	1
-2134084888	29	23	9	14	0	6
-214665832	21	21	7	14	0	0
-337748765	8	8	8	0	0	0
-655808311	10	8	6	2	0	2
-790646361	21	17	9	8	0	4
1405331961	10	10	8	2	0	0
1470713712	10	9	1	8	0	1
1482174936	14	12	12	0	0	2
603259629	15	15	11	4	0	0
781521487	16	15	8	7	0	1
Total	217	197	119	77	1	20

The local release candidate's 651-row derived ledger identifies each model/item, original/offline/final validity, label, origin, final failure and cluster. Frozen raw responses, prompts, request configurations and original/recovery hashes support independent verification. Prepared local release copies redact account/project and upload-location

metadata while preserving scientific content and recording every transformed hash. No human-verification ledger contains accepted labels. The audit supplies no semantic precision, recall, model superiority, autonomous-system validation or approved training data.

S6.3 Independent Reference Protocol: Prepared but Uncompleted

A separate first-round protocol specifies 36 probability-sampled and 12 targeted relationships, with two overlaps: 46 distinct cases across 16 problems. It has not produced independently accepted reference labels. Its sampling frame is the selected 217-item collection, not the whole NL4Opt corpus. Independent source-reference and template review would be necessary before any future use as supervision. The completed D1 author audit is a different construction and does not fulfill this protocol.

S7 Preserved Core-Results Definitions and Archival Details

S7.1 Structural and Solver-Backed Validation

Core metrics are solver-free. Three historical restricted studies use a different grounding implementation: optimization-role relation repair, rather than the headline typed-greedy configuration. The 60-case study takes the first original queries with 2–8 gold scalar parameters; the 269-case study selects instances with reference solver code. Their saved aggregate rates are retained, but their names must not be interpreted as stronger validation than the code performs.

The legacy field `structural_valid` is a *coverage/type proxy*: it is true when the predicted eligible inventory is empty, or when coverage and type match are both at least 0.8. Unlike main-benchmark readiness, this proxy treats an empty inventory as passing. It does not inspect linearity, decision variables, objective correctness or constraint well-formedness. The field `parseable_model_artifact` checks only that objective and constraint metadata are nonempty. `instantiation_complete` combines that presence check, full predicted-scalar coverage, and absence of expected parameter names in the rendered description; it does not require compatible types or validate numerical values. These subset inventories use numeric status in the predicted schema's parameter record, not the main benchmark's intersection with the query's gold keys.

The 20-case solver study is a separate, compatibility-filtered execution check. Its executable flag means that reference code finishes without an exception and leaves a model object. The two legacy success/feasibility flags both test the shim's status 2, set after a successful SciPy optimization with a returned solution; they are not independent outcomes. The objective-produced flag checks for a non-null objective value. Predicted-template parameter records initialize execution: unfilled scalars and structured data retain reference values, and filled parameters originally typed as integers are rounded. Thus these are gold-assisted executions of reference programs, not complete programs generated from the query alone.

Coverage/type proxy subset (60 instances). Table S5 retains the historical schema-hit, coverage/type-proxy and template-completion rates. Neither proxy tests numerical or formulation correctness.

Extended proxy subset (269 instances). On a larger subset with available reference solver code, the historical report records for TF–IDF, BM25, and Oracle schema-hit rates of 0.9368, 0.9257, and 1.0000; coverage/type-proxy rates of 0.8141, 0.8104, and 0.8253; and template-completion rates of 0.6654, 0.6580, and 0.6840. Solver execution was not evaluated here because the reference path required Gurobi support unavailable in the evaluation environment; executable conclusions are reserved for the HiGHS-compatible subset below.

Restricted solver-backed subset (20 instances). The historical script selects the first 20 reference programs passing a literal substring compatibility filter for the shim. It constructs one subset before either condition, but the original ID manifest is unavailable, preventing independent membership or paired-outcome checks. Table S6 retains its reported rates. The 0.80 versus 0.75 solver-success gap is one descriptive case, not a tested improvement. The report attributes non-executions to two shim `IndexError`s among 40 method–instance evaluations; original logs are unavailable.

The extended report records 69 missing-slot, 82 type-mismatch and 267 template-incompletion events across three baselines (807 method–instance evaluations). These categories can overlap and are not a partition of 269 unique queries. The completed archival investigation found no original subset manifests or outcome files. Case examples in the 60/269 reports establish some shared membership, not full samples; these studies must not be treated as disjoint. The complete rates remain aggregate-supported. The main numerical conclusions do not rely on these proxy checks.

S7.2 Computational Complexity and Runtime

The recorded lexical backbone has a small computational footprint on this catalog. TF–IDF fits a sparse vectorizer once over the $M = 335$ schema documents and, for each query, transforms the query and scores it against the full catalog matrix by sparse cosine similarity (SCIKIT-LEARN); with fixed small M, the lexical query path consists of sparse vector operations and does not execute a neural encoder. Rule-based extraction is linear in query length. Typed-greedy assignment scans candidates in $O(|\mathcal{P}| \cdot |\mathcal{T}|)$ comparisons in the (typically small) numbers of eligible slots and mentions. A historical timing summary for TF–IDF + ratio-aware typed greedy (331 queries, `PYTHONHASHSEED=0`) records wall time 1.09 s (3.29 ms/query) with peak RSS $\approx$ 198 MB. The recorded scope excludes BGE-M3 embedding load/compute and remote LLM baselines; the single retained timing summary is contextual, not a reproducible timing distribution or cross-method efficiency benchmark.

S7.3 Challenges in Numerical Mention Grounding

The stored-value failures motivate an occurrence-level relationship audit; they do not by themselves identify semantic error causes. A mention has a source occurrence, not just a magnitude: for the auxiliary relation audit, write $m = (a, b, v)$ for its half-open

character interval and normalized value, and $r = (q, m, p)$ for the proposed relation to parameter p. This diagnostic representation does not imply that the historical typed-greedy outputs retained character-aligned assignment traces. Two occurrences with the same v can refer to different resources; one occurrence may also legitimately support several declarations. No-reuse assignment is therefore an experimental restriction, not a general law of optimization modeling. Implicit objective or identity terms introduce a different problem: there may be no lexical number to align. Their absence cannot be converted automatically into a verified null relationship.

Historical weak-label learning used numerical equality to choose a first matching mention. Such targets do not distinguish repeated values, resource roles or incomplete formulations. Earlier neural and classifier-plus-assignment runs are documented, but their supervision and evaluation do not support a clean conclusion that learning cannot improve grounding. The saved P0 comparison shows no coarse-readiness gain from its exact-assignment condition; missing prediction traces and evaluation-derived splits prevent a broader semantic interpretation. Online Resource 1, Section S3.6 distinguishes completed negative results, inconclusive runs and blocked proposals.

S7.4 Preserved Statistical and Robustness Audit History

All inferential analyses here are retrospective. Query identities are paired across methods; the correction-only reanalysis reuses the existing 327 delexicalized text groups for all 331 original queries. It reports marginal paired group-percentile bootstrap intervals (B = 10,000, NumPy seed 42), exact groupwise method-swap tests on integer count differences, and Holm adjustment over the 16 reported retrieval/readiness contrasts [2]. That family contains five main-table contrasts, the BGE–TF–IDF retrieval contrast, and ten ordinary/Strict contrasts against the five grounding alternatives. The factorial's separate 240-endpoint family remains unchanged. Grouping is heuristic and does not remove all dependence or prior benchmark exposure.

Table 6 replaces the legacy significance summaries using the same saved observations. The earlier uncentered bootstrap sign-tail values were not null-calibrated tests; the BGE intervals were shifted randomization quantiles rather than sampling confidence intervals. Floating-point ties also affected those BGE p-values. Historical statistic files are retained in the correction record. The ratio-aware extraction gain remains $8/331 = 0.0242$; its exact paired/group test is $p = 0.0078125$ and the adjusted value is 0.0302. These tests concern coverage/type diagnostics, not numerical correctness. The historical sanitized-retrieval aggregates (Table S4) report preserved or improved rates after stripping numbers and/or stopwords. Without the original per-condition predictions, this is limited support for robustness to that transformation, not independent confirmation of its mechanism. The diagnostic uses a closely matched index (331 documents excluding four filler templates; no `doc_id` tokens), so TF–IDF/BM25 baseline cells differ slightly from Table 1 while LSA matches exactly (0.8459). A Jaccard-overlap stratification places 327/331 queries in a high-overlap bucket (TF–IDF R@1 0.9174); the medium-overlap bucket has only 4 queries and is reported only as a qualitative caveat.

To check whether wrong schemas can inflate ordinary InstantiationReady via smaller eligible-slot sets, a repository diagnostic of selective schema reranking raised ordinary readiness from 257/331 to 265/331, but 6 of 8 newly ready queries used an incorrect

schema. We therefore use the schema-gated readiness measure as a robustness check (Table 7). The gate removes 10 TF–IDF queries in both extraction configurations and widens the descriptive TF–IDF–Oracle readiness gap from 0.0483 to 0.0785. The corrected ordinary/Strict adjusted p-values are 0.0028 and < 0.001, respectively; neither metric tests scalar values.

S8 Additional Completed Investigations and Evidence Status

The accompanying evidence register, `docs/jdsa_revision_phase1_20261029/EXPERIMENT_EVIDENCE_REGISTER.md`, assigns 47 investigation groups to explicit publication destinations. These are not 47 independent samples or 47 completed performance experiments. The register distinguishes completed, partial, inconclusive, blocked and proposed work, identifies source hashes and commits, and separates saved-prediction verification from metric reaggregation and summary-only evidence. The following additions preserve relevant work previously absent or only briefly mentioned; they do not replace the main matched comparison. No experiment, prediction, model call or solver run was performed for this revision.

S8.1 Earlier Grounding Campaigns and Baseline-Version Corrections

The archived ESWA-era campaign evaluates 20 configurations over the same 331-query frame in three variants. Table S19 reaggregates ordinary readiness from its preserved per-query CSVs. The historical typed baseline has 175 ready original queries, not the 174 stated in one older narrative summary; the retained row observations take precedence. These are historical configurations, not new evaluations of current code. Coverage, TypeMatch, conditional Exact20 and ordered partitions from all 60 files remain in the accompanying offline check record.

Global-compatibility local/pairwise/full, relation-aware basic/operators/semantic/full, and ambiguity candidate/beam/abstention/full therefore remain represented, including their negative readiness observations. The ambiguity-abstention configuration has original-query coverage 0.2207. These measurements do not establish semantic error causes, calibrated confidence, or a general inability of richer grounding to improve complete numerical agreement. Older bootstrap sign-tail significance claims are not reused.

A later same-version audit (Table S20) corrected a misleading comparison against stale typed-greedy readiness 0.5287. Maximum-weight matching's 0.7432 was lower than the audit's typed-greedy 0.7764, not evidence of an improvement over that contemporaneous baseline. Its paired transitions contain seven rescues and 18 regressions. Search-structured and hierarchical-structured grounding both have 233 ready queries, without establishing that their assignments are identical. This audit predates the ratio-aware configuration; its Oracle and constrained values must not replace the later main-table values. The main paper's improved all-key counts remain a different, value-aware finding.

Table S19 Ordinary-ready counts in the retained legacy campaign; denominator 331 for every cell. These input-corruption and scoring variants are distinct from the ratio-aware main comparison. No historical significance claim is inferred from this descriptive table.

Historical method	Orig	Noisy	Short
Typed greedy	175	13	5
BM25 + typed	172	14	5
LSA + typed	168	14	4
Oracle + typed	188	14	5
Constrained	140	12	5
Role repair	146	0	2
Semantic IR	161	0	2
Acceptance rerank	174	14	7
Hierarchical rerank	172	14	7
Global compatibility: local	138	0	2
Global compatibility: pairwise	140	0	2
Global compatibility: full	143	0	2
Relation-aware: basic	165	0	2
Relation-aware: operators	155	0	2
Relation-aware: semantic	144	0	2
Relation-aware: full	138	0	2
Ambiguity: candidate greedy	140	0	2
Ambiguity: beam	140	0	2
Ambiguity: abstain	9	0	1
Ambiguity: full	139	0	2

S8.2 Ranked Retrieval, Reranking and Diagnostic Projections

Historical higher-rank retrieval aggregates are preserved in Table S21. Main top-1 rates can be checked against retained schema IDs; the full original ranked lists needed to independently recalculate every higher-rank cell were not authenticated. The values below therefore retain aggregate-supported status and are not independent replications of retrieval.

The top-k diagnostic locates the 30 TF–IDF top-1 misses at ranks 2, 3, 4–5, 6–10 and beyond 10 in counts 9, 5, 4, 6 and 6. Thus 325 targets occur in the top ten. A gold-schema top-ten oracle yields 270 ordinary-ready queries, while a readiness-maximizing oracle yields 286, including 32 wrong-schema-ready cases. Among tested heuristic rules, the largest observed readiness count is 282, with 20 schema regressions. These are exposed diagnostic ceilings/rule comparisons, not deployable oracle performance or new held-out validation. The subsequent selective-reranking result is described in main Section 5.6.

Table S20 August 12 baseline-staleness audit, saved ordinary-ready counts on 331 original queries per row. This is a separate historical configuration cohort.

Method	Ready /331
Typed greedy	257
BM25 + typed	253
LSA + typed	243
Oracle + typed	273
Constrained	248
Role repair	244
Semantic IR	237
Acceptance rerank	253
Hierarchical rerank	238
Maximum-weight matching	246
Search structured	233
Hierarchical structured	233

Table S21 Archived ranked-retrieval summaries, 331 queries in each variant. All cells are historical aggregates; higher-rank metrics do not measure scalar correctness.

Variant	Method	P@5	MRR@10	nDCG@10	Coverage@10
orig	BM25	0.9668	0.9182	0.9325	0.9758
orig	TF–IDF	0.9637	0.9334	0.9451	0.9819
orig	LSA	0.9456	0.8904	0.9112	0.9758
noisy	BM25	0.9728	0.9245	0.9386	0.9819
noisy	TF–IDF	0.9668	0.9315	0.9438	0.9819
noisy	LSA	0.9668	0.9240	0.9370	0.9758
short	BM25	0.8792	0.8178	0.8454	0.9335
short	TF–IDF	0.8731	0.8202	0.8436	0.9184
short	LSA	0.8761	0.8096	0.8342	0.9124

The role/quantity Stage-A audit examines 49 targeted wrong assignments on a 331-query frame: 28 are marked separable, 10 ambiguous and 11 not separable by its rules. It projects zero additional coarse-ready queries (257 before and after); its recorded all-query cascade has false-trigger rate 1.0. Readiness neutrality does not show that a corrected value would be scientifically irrelevant. The separate strict-failure precursor identifies 54 schema-hit/non-ready cases and 58 Oracle non-ready cases. Its ratio-word prototype moves Strict from 247 to 255 with eight gains and no losses, later

confirmed by the completed extraction ablation. Oracle projections of 254 for perfect extraction, 271 for expected-type repair, 272 for reuse and 247 for candidate filtering are not executed repair-policy outcomes.

S8.3 Candidate Availability and Full Pilot Preservation

Table S22 retains the completed candidate availability analysis over 331 queries and 1,847 reference scalar slots. The cache contains 311 queries/1,705 rows; missing selections are not dropped from the frame. These numerical product-of-pools ceilings allow reuse and do not certify source-occurrence identity or a selected complete mapping. Adding the coarse-type requirement reduces the top-eight whole-query tolerance ceiling from 199 to 139.

Table S22 Archived candidate-pool ceilings, not achieved inference accuracy. Slot denominators 1847; query denominators 331.

Pool	Exact slots	Tolerance slots	All exact queries	All tolerance queries
1	773	823	22	23
3	1227	1256	87	95
5	1379	1404	146	155
8	1456	1464	195	199
all	1467	1473	203	206

For the original 50-query/280-slot pilot, Table S23 preserves all six fixed configurations. Independent and joint methods use identical cached pools and inventory; the explicit-constraint branch activates zero supported parameter-only relations, so its duplicate outcome is a no-op, not evidence against useful mathematical constraints. On the overlap-free 49-query slice the strongest local and joint variants still have three full-tolerance successes. The historical pre-ratio cached choice is a separate configuration with 69 exact/80 tolerance-correct slots and two exact/tolerance-complete instances on the 50 IDs; it does not replace the headline ratio-aware result.

The initial cached error accounting over 1,847 gold slots records 167 wrong-schema slots, 1,053 acceptable-value-present but wrongly selected, 26 acceptable-value-present but abstained, 394 correct selections, 155 without an acceptable value in the pool, 19 without a saved pool, 29 without standalone metadata, and four omitted by metadata. These sum to 1,847. Candidate presence does not prove that the mention remained available after earlier assignments or expressed the right role. Section S3.2 reconciles the 394 metadata-filtered count with 401 when actual saved selections outside that later inventory are retained. Local/joint slot changes and all per-case outcomes remain in the original pilot artifacts; no later result substitutes missing historical predictions.

Table S23 Saved original 50-query joint pilot. E/T denote numerical exact/tolerance counts; slot denominator 280 and full-instance denominator 50. Full instances additionally require all gold keys and compatible types.

Configuration	E slots	T slots	E full	T full
Independent type	52	57	1	1
Independent text/roles	128	139	1	2
Joint text/roles	119	130	1	2
Independent text + units/types	134	144	2	3
Joint text + units/types	126	137	2	3
Joint + explicit constraints	126	137	2	3

S8.4 Additional Learned-Model Evidence and Small Auxiliary Runs

Section S3.6 preserves the central supervision findings. The earlier 200-step neural run used a reported 14,298-pair file and a test-as-train fallback; it establishes execution, not a valid held-out benchmark. The 184-step run on 35/7/8 problems records 5/59 slot selections and 0/8 legacy exact instances. The largest 230/50/50 run records 51/259 aligned-slot decisions, 51/280 overall, 0/45 legacy exact and 19/280 type matches. Its purported rule comparator conflicts with a later summary; neither is promoted into a clean causal learning comparison.

P0's development ablation records 115/245 correct slot groups for logistic regression on combined features, 114/245 for boosted trees, 109/245 for engineered features alone and 58/245 for embedding features alone. Its 50-case coarse-readiness test results remain .80/.78/.78 for independent/exact/repair decoding. The GAMS auxiliary run uses 17 models, 1,135 pairs and 85 heuristic positives but only two effective test problems: baseline 2/11 versus auxiliary 1/11 slot decisions, 0/2 exact for both. Targeted synthetic auxiliary training uses 155 pairs and one test problem: 0/8 versus 1/8 slot decisions, 0/1 exact both, and type matches 7/8 versus 0/8. These tiny, weakly supervised observations support neither broad transfer gains nor broad negative claims.

Separate historical MiniLM schema-retrieval training has retained completed fitting records (including an 80,297-pair run), but addresses retrieval rather than occurrence binding. No completed non-smoke MS-MARCO MiniLM cross-encoder or NL4Opt auxiliary neural checkpoint was authenticated. Implementations, export stubs and proposed clean external-supervision protocols are not counted as model outcomes.

S8.5 Archived Two-Stage LLM Scalar Baselines

The repository also retains older LLM schema-selection/slot-filling measurements (Table S24). These differ from both the common-18 program-generation study and the 217-item relationship audit. Rows are reaggregated from saved query metrics, not regenerated assignments. The historical provider tags do not by themselves authenticate exact model snapshots, complete prompts or serving defaults. Supplied catalogs/slot

inventories and incomplete protocol lineage prevent treating these as autonomous or harmonized provider comparisons. The short-input row illustrates why high conditional Exact20 can coexist with very low readiness.

Table S24 Archived two-stage scalar-baseline observations. Each row has 331 queries. Exact20 uses the explicitly stated comparable schema-hit denominator; no provider ranking or semantic accuracy is inferred.

Recorded tag	Variant	Schema hits	Ready	Exact20	Comparable n
openai	orig	307	280	0.8291	294
openai	noisy	309	116	0.1461	171
openai	short	243	4	0.8730	63
mistral	orig	171	154	0.9101	157

The optional Gemini scalar rerun has infrastructure/preflight records but no authenticated completed full-benchmark artifact in the inspected rerun directory. This does not negate the separate completed Gemini relationship study. The proposed CloudRift Qwen extension stopped before inference at its historical availability gate: all 217 targets were unattempted, not invalid judgments or zero-accuracy observations. No fourth-provider result is reported.

S8.6 Semantic Contrast Construction and Unexecuted Studies

The completed D1 construction and all 12 author-validated pair summaries are preserved in Section S9.2. Its assisted annotation and shared construction lineage preclude independently blinded reference or provider-performance claims. The separate independent review protocol for the 217 relationships remains uncompleted (Section S6.3).

An earlier similarly named global-consistency implementation has only a synthetic catalog-derived report (331 cases; typed readiness 16 versus global 18 on original text); real-data evaluation was blocked. Its temporary harness and original outputs are not authenticated here. Those artificial-reference figures are retained as historical context, excluded from the scientific real-data comparison, and not confused with the completed global-compatibility family in Table S19. DeepOR and OR-R1 have no evaluated common-18 generations; their availability context remains in Section 5.8. G1–G3 are closed feasibility/identifiability findings, not zero-performance experiments. These statuses preserve the research record without fabricating observations.

S9 External Extraction and Semantic Contrast Details

S9.1 OptMATH Numeric Extraction

As a component-level check, we evaluated numeric extraction on a fixed 1,000-instance OptMATH-Train sample [5] (no fixed schema catalog, so extraction only). Pooled raw AST code-value overlap recall is 0.7501; pooled automatic text-support recall is

0.8118 (complete-instance rate 0.5910). The saved automatic category counts classify 2,310/31,736 = 7.28% of code-side values across the 1,000 instances as implementation, solver/API, index/loop or array/dimension constants. This is distinct from the 150-instance classifier comparison (seed 20260815). The later verification summary reports audit-calibrated text-supported recall as the pooled (micro) ratio 23,241/28,799 = 0.8070 (95% instance-bootstrap CI $[0.7842, 0.8283]$, $B = 10{,}000$, seed 42, over the 1,000 sampled instances). Its corrected per-instance counts are unavailable for independent rebootstrap of that historical CI. The reported mean of per-instance ratios (macro recall) is 0.8135. The stored verification JSON reverses the names of these two aggregates; the values are preserved here with their implemented definitions. The 28,799 values are not independent samples, and occurrence-aware recall is not identifiable from these set-based artifacts. Two independently coded classifiers agree on 98.04% of 4,801 literals in the 150-instance sample—automated cross-validation, not human gold labels. This is limited evidence that extraction transfers beyond NLP4LP; it does not establish full-framework generalization.

S9.2 D1 Construction and Interpretation

The D1 task fixes a proposed relationship and asks whether it expresses a marked source fact in full context. Each pair changes a controlled phrase or supplies a meaning-preserving paraphrase. Tables S25 and S26 summarize all 12 accepted pairs, retaining the final author-reviewed labels rather than treating an earlier generated formalization as authoritative. C and I denote CORRECT and INCORRECT relative to the fixed candidate; they are author-validated, language-model-assisted labels. The completed construction contains 24 samples, 12 C and 12 I labels, ten meaning-changing pairs and two paraphrase pairs across eight categories. No provider predictions or performance measurements exist for this set.

The final audit accepts all 12 pairs for descriptive use, with none excluded or unresolved. Acceptance is scoped to the marked fact and supplied candidate. Pair 03-01 does not require a bounded complete optimization problem to distinguish objective directions. Pair 04-03 rejects the fixed 24-minutes-per-pack candidate in the base text, but does not certify the generated $3x$ expression for arbitrary partially filled crates. Pairs 03-03 and 08-03 do not isolate a causal effect of role independently of unit changes.

Twenty final labels were directly exposed to language-model assistance: one guided revision and 19 later decisions. Twelve final decisions were batch-approved. Four retained first-pass labels are not certified independently blinded labels. All cases share a generated-fixture lineage; ten archived broad template groups are not ten independently certified families. The negative bound, prohibition and conditional cases require their explicit domain assumptions. Private author-review conversations are not included in the public manuscript package. These dependencies prevent treating the set as independent human gold, as a provider benchmark with observed accuracy, or as an explanation of the core pipeline's measured failure frequencies.

Table S25 D1 contrast pairs, part 1. IDs omit the common D1P prefix. Labels are base/contrast judgments about the fixed candidate, not judgments about a generated complete model.

Pair	Category	Source contrast and fixed candidate	Labels
01-01	Negation	Not fewer than seven cabinets versus not more than seven; candidate $x \geq 7$. A shared boundary does not imply equivalence.	C/I
01-06	Negation	No mercury drum may be loaded versus loading any drum is prohibited; candidate $x \geq 1$ contradicts both nonnegative count requirements.	I/I
02-01	Bound direction	At most 19 shelves versus at least 19; candidate $x \leq 19$.	C/I
02-04	Bound direction	Signed inventory change at least -4 versus at most -4 tonnes; candidate $d \leq -4$. No nonnegativity is implicit for d.	I/C
03-01	Objective/ constraint roles	Minimize versus maximize route distance x; candidate $\min x$. This tests direction, not a change from objective to constraint.	C/I
03-03	Objective/ constraint roles	Nine dollars profit per bicycle versus nine minutes labour per bicycle; candidate assigns 9 to labour in minutes. Role and units change together.	I/C

Table S26 D1 contrast pairs, part 2. Together with Table S25, these cover eight categories. Pairs 01-06 and 07-05 preserve meaning; the remaining ten change it.

Pair	Category	Source contrast and fixed candidate	Labels
04-01	Units	Fifteen minutes versus fifteen hours per chamber; candidate 1/4 hour per chamber.	C/I
04-03	Units	Twenty-four minutes per eight-pack crate versus per pack; candidate 24 minutes per pack. A full-crate average does not establish a global linear rule for partial crates.	I/C
05-01	Variable linkage	East tank x versus west tank y has 41-litre capacity; candidate $x \leq 41$. Capacity is not equality.	C/I
06-03	Ratios	Coastal budget x is 20% more than versus 20% of inland budget y; candidate $x - 0.2y = 0$. Shared zero budgets do not imply equivalent relations.	I/C
07-05	Conditionals	Standby power a required unless grid b is selected versus if grid is not selected; candidate $\neg b \Rightarrow a$. Both permit $a = b = 1$.	C/C
08-03	Irrelevant information	Temperature 22 degrees, explicitly not price, versus price 22 dollars, explicitly not temperature; candidate $\min 22x$ for lantern purchases. Role and units change together.	I/C

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